# Vision-Only Latent Diffusion Super-Resolution without T2I Pretraining

Work-in-progress technical report

## Abstract

This project trains an x4 super-resolution model directly, without importing a pretrained text-to-image diffusion backbone. The active implementation is Stage 1: a factor-4 VAE / autoencoder trained on 512px HR crops so later stages can operate in latent space.

## Goal

- LR 128x128 -> HR 512x512.
- LR 192x192 -> HR 768x768 later.
- One future model for photo and anime/illustration domains through domain conditioning.
- PyTorch/ROCm first, no custom CUDA/ROCm ops.

## Planned Pipeline

HR image -> factor-4 VAE -> HR latent.
LR image -> condition encoder -> multi-scale LR features.
Noisy HR latent + LR features + timestep + domain embedding -> diffusion U-Net.
The VAE decoder maps the final denoised latent back to the x4 HR output.

## Current Implementation

- Manifest dataset loader with photo/anime domain IDs.
- On-the-fly x4 degradation pipeline.
- Factor-4 AutoencoderKL.
- bf16 PyTorch/ROCm training.
- W&B online logging.
- Fixed LR / GT / HR validation image logging.
- Validation metrics: loss, recon, KL, MSE, PSNR.

## Current Training Run

- Config: configs/autoencoder_photo10k.yaml.
- Run: autoencoder_photo10k_b16_eval_online.
- Training data: 10,000 photo images.
- Validation data: 100 photo images.
- HR crop: 512px.
- Latent: 128px, 16 channels.
- Batch size: 16.
- Max steps: 100,000, about 160 epochs.
- Eval cadence: every 1,000 steps.
- Fixed sample cadence: every 500 steps.

## Evaluation

The current evaluation is VAE-only: HR -> encode -> decode -> reconstruction. Fixed validation samples are logged so visual progress is comparable over time. The best checkpoint is selected by eval/recon, with PSNR and sample quality as supporting signals.

## Roadmap

1. Finish Stage 1 VAE training and choose a checkpoint.
2. Train deterministic LR -> HR latent pretraining.
3. Train conditional latent diffusion for SR.
4. Add perceptual/GAN fine-tuning if needed.
5. Distill to fewer inference steps.
6. Evaluate checkpoints with fixed A/B preference tests.

## Limitations

This is not yet a complete SR diffusion model. The current active run trains only the VAE reconstruction model. Anime/illustration data has not yet been added to the active training set, and this report does not claim a novel published method.